
Medico-Legal Platform

ADMIN DASHBOARD

COMPLETE DEVELOPMENT PROMPT

Master Engine • Appointment Engine • Finance Engine • Report Engine • CRM • Staff • Analytics
• Security

Internal Staff Operations System — Production-Ready
Version 1.0 • June 2025

HOW TO USE THIS PROMPT

This document is the single authoritative development prompt for the Kutlwano & Associates Admin Dashboard. Paste the full relevant section into your AI code assistant (Cursor, Copilot, Claude, GPT-4) BEFORE generating any file. Every rule here overrides any default the AI would otherwise apply. Read Section 1 (Core Rules) before anything else — those constraints never change regardless of which module you are building.

- Each section = one self-contained prompt block
- Always include Section 1 (Core Rules) at the top of every prompt
- Append the specific module section below it
- Engine sections must be built before their consumer modules

SECTION 1 — CORE RULES & NON-NEGOTIABLE CONSTRAINTS

1.1 CSS & Styling

The project has an existing raw CSS design system. This must be used as-is. You are writing production code that extends existing work — not starting fresh.

ABSOLUTE CSS RULES — THESE ARE HARD STOPS:

- ZERO Bootstrap, Tailwind, Foundation, or any CSS utility framework
- ZERO Google Fonts — system font stack only
- ZERO CDN-loaded CSS of any kind
- ZERO external icon libraries (FontAwesome, Material Icons, Heroicons, etc.)
- ZERO inline style overrides that conflict with existing class names
- ZERO UI component kits (Shadcn, Radix, Ant Design, MUI, Chakra, etc.)
- Icons must be inline SVG only, using viewBox, matching existing icon size
- All spacing, colour, and typography must use the existing CSS variables
- After splash screen → loading → the EXACT login page must appear. Do not alter the login page layout, form positioning, colours, or typography in any way.

1.2 Technology Stack

Layer	Technology	Notes
Frontend	React + TypeScript	Strict mode enabled
Backend / DB	Supabase	PostgreSQL + Realtime + Storage
Auth	Supabase Auth + Custom OTP	OTP via Resend email
Edge Functions	Supabase Edge Functions (Deno)	API layer, notifications, reports
File Storage	Supabase Storage	Documents, reports, images
Routing	React Router v6	Protected routes with role guards
State	React Context + useReducer	No Redux, no Zustand
Forms	Controlled components	No react-hook-form, no Formik
HTTP	Supabase JS Client v2	No Axios, no Fetch wrappers
Deployment	Render.com	Environment variables in Render dashboard

1.3 Data & Quality Rules

- ZERO mock data, seed data, or placeholder values anywhere in the codebase
- Every value displayed on screen comes from a real Supabase query
- Every form submission writes to the real Supabase database
- All loading states must be handled with skeleton loaders, not spinners
- All error states must render a user-readable error card with retry option
- All empty states must render a descriptive empty-state component
- No console.log left in committed code — use a structured logger utility

- TypeScript strict mode: no 'any', no 'unknown' without explicit narrowing
- All database calls go through the service layer — never from components directly

1.4 Security Rules

- All admin routes gated by role: only 'admin' and 'staff' roles permitted
- Row Level Security (RLS) must be active on every Supabase table
- No sensitive data stored in localStorage or sessionStorage
- All user actions that mutate data must be written to the audit_logs table
- Session expiry handled gracefully — expired sessions redirect to /login
- OTP tokens expire in 10 minutes. Expired OTP shows clear error, no retry flood

SECTION 2 — FILE & FOLDER STRUCTURE

Every file must be placed in the correct location. The structure below is the canonical layout for the Internal (Admin/Staff) system. Do not deviate.

```
src/
  admin/
    pages/
      AdminDashboardHome.tsx      ← master dashboard overview
      AppointmentManagement.tsx   ← appointment list + master record
      AppointmentDetail.tsx       ← single appointment full view
      AttorneyManagement.tsx
      ExpertManagement.tsx
      ClaimantManagement.tsx
      FinanceDashboard.tsx
      ReportsDashboard.tsx
      StaffManagement.tsx
      AuditLogs.tsx
      CommunicationsCenter.tsx
      SystemSettings.tsx
      Analytics.tsx
    components/
      layout/
        AdminLayout.tsx          ← root layout wrapper
        AdminSidebar.tsx
        AdminHeader.tsx
        AdminBreadcrumbs.tsx
      appointment/
        AppointmentCard.tsx
        AppointmentTable.tsx
        AppointmentStatusBadge.tsx
        AppointmentChecklist.tsx
        AppointmentTimeline.tsx
        AppointmentNotes.tsx
        AppointmentActions.tsx
      finance/
        InvoiceCard.tsx
        PaymentTable.tsx
        FinanceSummaryWidget.tsx
      reports/
        ReportCard.tsx
        ReportGeneratorModal.tsx
      crm/
        AttorneyCard.tsx
        ExpertCard.tsx
        ClaimantCard.tsx
      shared/
        StatCard.tsx
        DataTable.tsx
        SearchBar.tsx
        FilterBar.tsx
        ModalWrapper.tsx
```

```
■ ■ ■ ■■■ ConfirmDialog.tsx
■ ■ ■ ■■■ SkeletonLoader.tsx
■ ■ ■ ■■■ EmptyState.tsx
■ ■ ■ ■■■ ErrorCard.tsx
■ ■ ■ ■■■ NotificationPanel.tsx
■ ■■■ hooks/
■ ■■■ useAppointments.ts
■ ■■■ useFinance.ts
■ ■■■ useReports.ts
■ ■■■ useCRM.ts
■ ■■■ useStaff.ts
■ ■■■ useAuditLog.ts
■ ■■■ useNotifications.ts
■■■ engines/
■ ■■■ appointmentEngine.ts ← MASTER ENGINE
■ ■■■ financeEngine.ts
■ ■■■ reportEngine.ts
■ ■■■ communicationEngine.ts
■ ■■■ documentEngine.ts
■ ■■■ notificationEngine.ts
■ ■■■ auditEngine.ts
■ ■■■ workflowEngine.ts
■■■ services/
■ ■■■ appointmentService.ts
■ ■■■ financeService.ts
■ ■■■ reportService.ts
■ ■■■ communicationService.ts
■ ■■■ documentService.ts
■ ■■■ notificationService.ts
■ ■■■ auditService.ts
■ ■■■ staffService.ts
■ ■■■ attorneyService.ts
■ ■■■ expertService.ts
■ ■■■ claimantService.ts
■■■ types/
■ ■■■ appointment.types.ts
■ ■■■ finance.types.ts
■ ■■■ report.types.ts
■ ■■■ user.types.ts
■ ■■■ crm.types.ts
■ ■■■ system.types.ts
■■■ utils/
■ ■■■ formatters.ts
■ ■■■ validators.ts
■ ■■■ dateUtils.ts
■ ■■■ logger.ts
■ ■■■ roleGuard.ts
```

SECTION 3 — MASTER ENGINE FILES (CORE BUSINESS LOGIC)

Engine files are the single source of truth for all business logic. They are pure TypeScript — no React, no JSX, no UI logic. They are imported by service files. Components never call engines directly.

3.1 appointmentEngine.ts — THE MASTER ENGINE

This is the most critical file in the entire platform. Every appointment record, status transition, checklist evaluation, expert assignment, financial trigger, report trigger, and communication trigger originates here. Build this file first before any other engine.

Appointment Lifecycle States:

State Key	Label	Who Triggers	Next States
DRAFT	Draft	Attorney (portal)	PENDING_APPROVAL
PENDING_APPROVAL	Awaiting Approval	System (auto)	APPROVED, REJECTED
APPROVED	Approved	Admin Staff	SCHEDULED
REJECTED	Rejected	Admin Staff	DRAFT (resubmit)
SCHEDULED	Scheduled	Admin Staff	CONFIRMED, CANCELLED
CONFIRMED	Confirmed	Expert (portal)	IN_PROGRESS
IN_PROGRESS	In Progress	System (auto on date)	ASSESSMENT_PENDING
ASSESSMENT_PENDING	Assessment Pending	System (auto)	ASSESSMENT_COMPLETE
ASSESSMENT_COMPLETE	Assessment Complete	Expert	REPORT_PENDING
REPORT_PENDING	Report Pending	System (auto)	REPORT_SUBMITTED
REPORT_SUBMITTED	Report Submitted	Expert	REPORT_REVIEWED
REPORT_REVIEWED	Report Reviewed	Admin	INVOICE_PENDING
INVOICE_PENDING	Invoice Pending	System (auto)	INVOICE_SENT
INVOICE_SENT	Invoice Sent	Finance	PAID, OVERDUE
PAID	Paid	Finance	ARCHIVED
OVERDUE	Overdue	System (auto)	PAID, ESCALATED
ESCALATED	Escalated	System (auto)	PAID, WRITTEN_OFF
ARCHIVED	Archived	System (auto)	— terminal —
CANCELLED	Cancelled	Admin / Attorney	— terminal —

appointmentEngine.ts must export:

- createAppointment(payload: CreateAppointmentPayload): Promise
- updateAppointmentStatus(id, newStatus, actorId, reason?): Promise
- validateStatusTransition(current, next): boolean — throws if invalid
- assignExpert(appointmentId, expertId, actorId): Promise
- evaluateChecklist(appointmentId): Promise

- `getAppointmentMasterRecord(id)`: Promise
- `triggerFinanceEvent(appointmentId, eventType)`: Promise
- `triggerReportEvent(appointmentId, eventType)`: Promise
- `triggerCommunicationEvent(appointmentId, eventType)`: Promise
- `archiveAppointment(id, actorId)`: Promise
- `rescheduleAppointment(id, newDateTime, actorId, reason)`: Promise
- `getAppointmentTimeline(id)`: Promise

Master Appointment Record (AppointmentMasterRecord type):

The Master Record is the single aggregated object that the admin dashboard uses to render the full Appointment Detail view. It joins all related tables.

- `id`, `reference_number` (KAM-YYYY-NNNN), `status`, `created_at`, `updated_at`
- `claimant`: { `id`, `full_name`, `id_number`, `contact`, `address`, `injury_description` }
- `attorney`: { `id`, `firm_name`, `attorney_name`, `contact`, `matter_reference` }
- `expert`: { `id`, `full_name`, `specialisation`, `practice_number`, `availability_confirmed` }
- `appointment_datetime`, `venue`, `appointment_type`, `duration_minutes`
- `checklist`: `ChecklistItem[]` (each with `id`, `label`, `required`, `completed`, `completed_at`, `completed_by`)
- `assessment`: { `submitted_at`, `findings`, `diagnosis_codes`, `expert_notes`, `attachments[]` }
- `report`: { `status`, `submitted_at`, `reviewed_at`, `download_url`, `version_history[]` }
- `finance`: { `invoice_number`, `amount`, `vat_amount`, `total`, `paid_at`, `payment_method`, `receipt_url` }
- `communications`: `CommunicationLog[]` (email, SMS, system messages)
- `notes`: `InternalNote[]` (staff-only notes with timestamps and author)
- `timeline`: `TimelineEvent[]` (chronological audit of every state change)
- `documents`: `Document[]` (all files attached to this appointment)

Checklist System:

Every appointment has a dynamic checklist generated at creation time based on `appointment_type`. The checklist drives the workflow — no appointment can progress to `ASSESSMENT_PENDING` unless all required checklist items are marked complete.

- Checklist templates stored in `checklist_templates` table keyed by `appointment_type`
- Items are copied to `appointment_checklist` at appointment creation
- `evaluateChecklist()` returns { `allRequired`: boolean, `percentComplete`: number, `blockers`: string[] }
- Checklist items can only be completed by authorised staff roles
- Each completion is timestamped and attributed to the completing user

3.2 financeEngine.ts

Handles all financial logic including invoice generation, payment recording, VAT calculation, receipt generation, overdue escalation, and financial reporting.

- `generateInvoice(appointmentId, feeSchedule)`: Promise
- `recordPayment(invoiceId, paymentDetails)`: Promise
- `calculateVAT(amount, vatRate)`: number — default 15% South African VAT
- `markOverdue(invoiceId)`: Promise — triggers notification

- `escalateDebt(invoiceId)`: Promise — changes status to ESCALATED
- `generateFinancialSummary(dateRange)`: Promise
- `getOutstandingInvoices()`: Promise
- `getFeeSchedule(expertId, appointmentType)`: Promise
- `applyDiscount(invoiceId, discountPercent, authorisedBy)`: Promise
- `voidInvoice(invoiceId, reason, authorisedBy)`: Promise

Finance Engine Rules: All monetary values stored as integers in cents. Display layer converts to Rands (ZAR). Never store floats for money. Every financial action writes an entry to `finance_audit_log`.

3.3 reportEngine.ts

Manages the full lifecycle of medico-legal reports: submission, versioning, review, approval, and delivery. Reports are stored in Supabase Storage.

- `submitReport(appointmentId, expertId, fileBlob)`: Promise
- `reviewReport(reportId, adminId, reviewNotes)`: Promise
- `approveReport(reportId, adminId)`: Promise
- `requestRevision(reportId, adminId, revisionNotes)`: Promise
- `getReportHistory(appointmentId)`: Promise
- `generateReportCoverPage(appointmentId)`: Promise
- `deliverReportToAttorney(reportId)`: Promise — sends via email
- `generateReportsSummary(filters)`: Promise

3.4 communicationEngine.ts

All outbound and inbound communications are routed through this engine. It maintains a full log of every message, email, and notification.

- `sendEmail(to, templateId, variables, appointmentId?)`: Promise
- `sendOTP(email, purpose)`: Promise — via Resend
- `sendSMS(phoneNumber, message, appointmentId?)`: Promise
- `logInternalNote(appointmentId, authorId, note)`: Promise
- `getCommunicationLog(appointmentId)`: Promise
- `getEmailTemplates()`: Promise
- `updateEmailTemplate(id, content)`: Promise — admin only

EMAIL TEMPLATE IDs (must exist in `email_templates` table):

<code>APPT_CONFIRMATION_ATTORNEY</code>	— sent when appointment is APPROVED
<code>APPT_CONFIRMATION_EXPERT</code>	— sent when expert is assigned
<code>APPT_CONFIRMATION_CLAIMANT</code>	— sent 24h before appointment
<code>APPT_REMINDER_24H</code>	— automated 24h reminder
<code>APPT_REMINDER_1H</code>	— automated 1h reminder
<code>REPORT_SUBMITTED_NOTIFY</code>	— notify admin when expert submits report
<code>REPORT_READY_ATTORNEY</code>	— deliver report to attorney
<code>INVOICE_SENT</code>	— invoice delivery to attorney firm
<code>INVOICE_OVERDUE</code>	— overdue payment reminder

INVOICE_OVERDUE_FINAL	– final notice before escalation
PASSWORD_RESET	– OTP for password reset
ACCOUNT_CREATED	– welcome email for new portal users
APPOINTMENT_CANCELLED	– cancellation notification

3.5 auditEngine.ts

Every action taken by any user — admin, staff, attorney, expert, claimant — is written to the `audit_logs` table. This engine handles all write and query operations for the audit system. It is called by all other engines.

- `logAction(actorId, actionType, entityType, entityId, before, after, metadata?): Promise`
- `getAuditLog(filters: AuditFilters): Promise`
- `getEntityHistory(entityType, entityId): Promise`
- `getUserActivityLog(userId, dateRange): Promise`

AUDIT ACTION TYPES (constants):

```

AUTH: LOGIN, LOGOUT, OTP_SENT, OTP_VERIFIED, PASSWORD_CHANGED, ACCOUNT_LOCKED
APPOINTMENT: CREATED, STATUS_CHANGED, EXPERT_ASSIGNED, RESCHEDULED, CANCELLED, ARCHIVED
FINANCE: INVOICE_GENERATED, PAYMENT_RECORDED, INVOICE_VOIDED, DISCOUNT_APPLIED
REPORT: SUBMITTED, REVIEWED, APPROVED, REVISION_REQUESTED, DELIVERED
STAFF: CREATED, UPDATED, DEACTIVATED, ROLE_CHANGED
SYSTEM: SETTINGS_CHANGED, TEMPLATE_UPDATED, PERMISSION_CHANGED
CRM: ATTORNEY_CREATED, ATTORNEY_UPDATED, EXPERT_CREATED, EXPERT_UPDATED

```

3.6 notificationEngine.ts

- `createNotification(userId, type, title, body, link?): Promise`
- `markAsRead(notificationId): Promise`
- `markAllAsRead(userId): Promise`
- `getUserNotifications(userId): Promise`
- `getUnreadCount(userId): Promise`
- `subscribeToNotifications(userId, callback): Unsubscribe — Supabase realtime`

3.7 documentEngine.ts

- `uploadDocument(appointmentId, file, docType, uploadedBy): Promise`
- `getDocuments(appointmentId): Promise`
- `deleteDocument(documentId, actorId): Promise`
- `getSignedUrl(documentPath, expiresIn): Promise`
- `getDocumentsByType(appointmentId, docType): Promise`

DOCUMENT TYPES: REFERRAL_LETTER, ID_DOCUMENT, MEDICAL_RECORDS, ASSESSMENT_FORM, MEDICO_LEGAL_REPORT, INVOICE, RECEIPT, COURT_ORDER, CONSENT_FORM, OTHER

3.8 workflowEngine.ts

Orchestrates automated workflow transitions. Called by scheduled Edge Functions and triggered by events from other engines.

- runDailyWorkflow(): Promise — called by cron Edge Function
- checkOverdueInvoices(): Promise
- checkPendingAssessments(): Promise
- checkPendingReports(): Promise
- autoArchiveCompleted(): Promise — archives PAID appointments after 90 days
- sendScheduledReminders(): Promise

SECTION 4 — ADMIN DASHBOARD LAYOUT

4.1 AdminLayout.tsx

The root layout wrapper for all admin pages. It renders the sidebar, header, notification panel, and the main content area. Every admin page is wrapped in this component via the router.

- Left sidebar: fixed, 240px wide, collapsible on mobile
- Top header: fixed, 60px tall, spans the content area
- Content area: scrollable, starts at sidebar width + header height
- Notification panel: slide-in from the right, triggered from header bell icon
- All breakpoints handled in existing CSS — DO NOT add media queries

4.2 AdminSidebar.tsx

Navigation items and their exact routes:

Nav Item	Icon (SVG)	Route	Roles
Dashboard	grid-icon	/admin	admin, staff
Appointments	calendar-icon	/admin/appointments	admin, staff
Attorneys (CRM)	briefcase-icon	/admin/crm/attorneys	admin, staff
Medical Experts	stethoscope-icon	/admin/crm/experts	admin, staff
Claimants	users-icon	/admin/crm/claimants	admin, staff
Finance	currency-icon	/admin/finance	admin
Reports	document-icon	/admin/reports	admin, staff
Communications	mail-icon	/admin/communications	admin, staff
Analytics	chart-icon	/admin/analytics	admin
Staff	id-card-icon	/admin/staff	admin
Audit Logs	shield-icon	/admin/audit	admin
System Settings	cog-icon	/admin/settings	admin

4.3 AdminHeader.tsx

- Left: hamburger toggle (mobile sidebar) + current page breadcrumb
- Centre: global search bar — searches appointments by reference number
- Right: notification bell with unread count badge (realtime via Supabase)
- Right: logged-in user avatar + name + role label + logout option
- Notification count updates in real-time without page refresh

SECTION 5 — ADMIN DASHBOARD HOME (AdminDashboardHome.tsx)

The dashboard home is the first screen after login. It gives the admin or staff member a real-time operational overview of the entire platform. Every number is live from Supabase. No cached data, no mock totals.

5.1 KPI Stat Cards (Row 1)

8 stat cards in a responsive grid (4 per row on desktop, 2 on tablet, 1 on mobile):

- Total Appointments (all time)
- Appointments This Month
- Pending Approvals (clickable — links to filtered appointments list)
- Appointments Today
- Outstanding Invoices (count + ZAR total)
- Reports Awaiting Review
- Active Experts
- Active Attorneys

5.2 Quick Actions (Row 2)

- Log New Appointment (opens AppointmentCreateModal)
- Approve Pending (links to /admin/appointments?status=PENDING_APPROVAL)
- View Today's Schedule (links to /admin/appointments?date=today)
- Generate Report (opens ReportGeneratorModal)

5.3 Today's Appointments Table (Row 3)

- Columns: Reference, Claimant Name, Attorney, Expert, Time, Venue, Status, Actions
- Actions: View (→ /admin/appointments/:id), Quick Note, Change Status
- Empty state if no appointments today — show message with link to add

5.4 Pending Approvals Widget

- List of appointments with status = PENDING_APPROVAL
- Each row: Reference, Attorney Firm, Claimant Name, Submitted Date, Approve/Reject buttons
- Approve triggers appointmentEngine.updateAppointmentStatus → APPROVED
- Reject opens modal requesting rejection reason before confirming

5.5 Recent Activity Feed (Row 4)

- Last 20 audit log entries across the platform
- Format: [timestamp] [actor] [action] [entity] — e.g. 'Admin Sipho approved KAM-2025-0042'
- Rendered as a vertical feed with appropriate icons per action type
- Loads in real-time via Supabase realtime subscription

SECTION 6 — APPOINTMENT MANAGEMENT

6.1 AppointmentManagement.tsx (List View)

Full-featured appointment list with search, filter, and bulk actions:

- Search: by reference number, claimant name, attorney name, expert name
- Filter: by status, date range, expert, attorney, appointment type
- Sort: by date (asc/desc), status, reference number
- Pagination: 25 records per page with page controls
- Export: CSV export of filtered results (admin only)

Table columns:

- Reference No (KAM-YYYY-NNNN) — clickable → detail view
- Claimant Name
- Attorney / Firm
- Assigned Expert
- Appointment Date & Time
- Type (e.g. Orthopaedic Assessment, Psychiatric, etc.)
- Status Badge (colour-coded per status)
- Checklist % complete
- Actions: View | Assign Expert | Change Status | Cancel

6.2 AppointmentDetail.tsx (Master Record View)

The most important page in the admin system. Shows the complete AppointmentMasterRecord in a tabbed interface. Everything about this appointment is visible and actionable here.

Tabs:

- OVERVIEW — Reference, status badge, quick-action buttons, summary cards
- CLAIMANT — Full claimant profile, injury description, contact details
- ATTORNEY — Attorney and firm details, matter reference, contact
- EXPERT — Assigned expert, specialisation, availability confirmation status
- CHECKLIST — Checklist items with completion status, mark-complete buttons
- ASSESSMENT — Assessment form results submitted by expert
- REPORTS — Report versions, review status, approve/revision-request buttons
- FINANCE — Invoice, payment status, VAT breakdown, payment history
- COMMUNICATIONS — Full communication log (emails, SMS, system notes)
- DOCUMENTS — All attached documents with download/delete actions
- NOTES — Internal staff notes (not visible to external users)
- TIMELINE — Chronological audit trail of every change to this appointment

Status Change Rules (enforced on UI + engine):

- Status buttons only show valid next states — invalid transitions never shown

- Every status change requires a confirmation dialog
- REJECTED and CANCELLED always require a reason (mandatory text field)
- Status changes write to audit_logs with before/after values

6.3 Expert Assignment

- Opens a modal listing available experts filtered by appointment_type specialisation
- Shows expert availability status (available / busy on this date)
- Assignment triggers: expert notification email, appointment status update, audit log
- One expert per appointment — reassignment replaces current expert
- Reassignment requires reason and notifies the replaced expert

SECTION 7 — CRM MODULES

7.1 Attorney Management (AttorneyManagement.tsx)

List view with search and filter. Each card/row links to Attorney Profile.

- Fields: firm_name, attorney_name, email, phone, address, practice_number
- Stats per attorney: total appointments, active appointments, outstanding invoices
- Portal access status: active / inactive / pending OTP setup
- Actions: View Profile, Edit, Activate Portal, Deactivate, Send Welcome Email
- Attorney Profile: full details + linked appointments table + communication log

7.2 Medical Expert Management (ExpertManagement.tsx)

Expert directory with specialisation, availability, and performance stats.

- Fields: full_name, specialisation, practice_number, hpcsa_number, email, phone
- Regions/provinces they operate in — used for scheduling
- Fee schedule per appointment type (managed in expert fee schedule table)
- Stats: total completed assessments, reports submitted on time, avg turnaround
- Availability calendar view (read-only from admin — expert updates via portal)
- Actions: View Profile, Edit, Assign to Appointment, Deactivate, View Reports

7.3 Claimant Management (ClaimantManagement.tsx)

Claimant registry with linked appointments and document store.

- Fields: full_name, id_number, dob, gender, email, phone, address
- Injury description, linked attorney, legal matter reference
- All appointments for this claimant — chronological list
- Document history: all documents uploaded across all appointments
- Claimants do NOT get portal access by default — admin controls this

SECTION 8 — FINANCE DASHBOARD

8.1 Finance Overview

- Total invoiced (this month / this year)
- Total collected (this month / this year)
- Outstanding (overdue + pending)
- Overdue amount with count of overdue invoices
- Revenue trend chart — monthly bar chart (12 months rolling, ZAR values)

8.2 Invoice Management

- List of all invoices with: reference, appointment, attorney, amount, status, due date
- Status: DRAFT, SENT, PAID, OVERDUE, VOIDED, ESCALATED
- Actions: View, Send to Attorney, Mark Paid, Void, Apply Discount
- Generate Invoice: uses `financeEngine.generateInvoice()` with expert fee schedule
- Payment recording: `payment_method` (EFT, credit card), reference, date, amount

8.3 Finance Rules

RULES — NON-NEGOTIABLE:

- All amounts in ZAR (South African Rand)
- VAT at 15% applied to all invoices
- Monetary values stored as integers (cents) in the database
- Discount requires admin approval — captured in `finance_audit_log`
- Voiding an invoice requires reason and cannot be reversed
- All finance actions written to `finance_audit_log` AND `audit_logs`

SECTION 9 — REPORTS & ANALYTICS

9.1 Reports Dashboard (ReportsDashboard.tsx)

- Pending review: reports submitted by experts awaiting admin review
- Reports this month: total submitted vs approved vs revision requested
- Filter by: expert, appointment type, status, date range
- Each report row: appointment ref, claimant, expert, submitted date, status, actions
- Actions: View Report (opens signed Supabase URL), Approve, Request Revision, Deliver to Attorney

9.2 Analytics (Analytics.tsx)

- Appointments by status (pie/donut chart)
- Appointments by type (bar chart)
- Appointments by month (line chart, 12 months)
- Top 5 referring attorneys by appointment volume
- Expert utilisation: appointments per expert this month
- Average turnaround time: appointment date → report delivered
- Finance: monthly revenue line chart with target line

Charts must be rendered using inline SVG or a lightweight chart library already in the project. No Chart.js CDN. No D3 CDN. Pure inline SVG preferred.

SECTION 10 — STAFF MANAGEMENT & SYSTEM SETTINGS

10.1 Staff Management (StaffManagement.tsx)

Admin-only. Manages internal staff accounts and their roles.

- Fields: full_name, email, role (admin | staff | finance | viewer), status
- Create Staff: generates Supabase auth user, sends welcome email with OTP
- Edit: change name, email, role
- Deactivate: disables Supabase auth, logs action — does NOT delete
- Staff cannot delete themselves or other admins
- Role permissions matrix displayed as reference on this page

10.2 Audit Logs (AuditLogs.tsx)

Admin-only. Read-only view of the full audit trail.

- Filter by: actor, action type, entity type, date range
- Columns: Timestamp, Actor, Action, Entity Type, Entity ID, Before, After
- Before/After values shown as expandable JSON diff
- Export to CSV (date-ranged)

10.3 System Settings (SystemSettings.tsx)

Admin-only. Platform configuration.

- Company details: name, address, VAT number, logo
- Email templates: view and edit all templates (rich text editor)
- Fee schedules: appointment type fee matrix per expert specialisation
- Checklist templates: manage checklist items per appointment type
- OTP settings: expiry time (default 10 min), max attempts
- Notification rules: which events trigger which notifications
- All settings changes written to audit_logs

SECTION 11 — DATABASE TABLES (SUPABASE / POSTGRESQL)

These tables must exist in Supabase. All tables have RLS enabled. Use snake_case for all column names. All timestamps are timestamptz (UTC). All primary keys are UUID generated by gen_random_uuid().

Table Name	Primary Purpose
profiles	Extends Supabase auth.users — stores role, name, status
appointments	Master appointment record — all core fields
appointment_checklist	Checklist items per appointment (copied from template)
checklist_templates	Template checklist items per appointment_type
assessments	Expert assessment submissions linked to appointment
reports	Report versions submitted by experts
invoices	Finance invoices linked to appointments
payments	Payment records linked to invoices
finance_audit_log	All financial events
attorneys	Attorney and firm records
experts	Medical expert profiles
claimants	Claimant records
expert_fee_schedules	Fee matrix: expert × appointment_type → amount (cents)
staff	Internal staff records
communications_log	All emails, SMS, OTPs sent/received
email_templates	Email template content keyed by template_id
notifications	In-app notifications per user
documents	Document metadata + Supabase Storage path
internal_notes	Staff-only notes on appointments
audit_logs	Full audit trail — all actions across all entities
appointment_timeline	Chronological events for a single appointment
system_settings	Key-value store for platform configuration

Naming Conventions:

- All tables: snake_case, plural (e.g. appointments, not Appointment)
- Foreign keys: referenced_table_id (e.g. appointment_id, attorney_id)
- Status columns: type TEXT with CHECK constraint listing valid states
- All tables have: created_at (default now()), updated_at (trigger-maintained)
- Soft deletes: use is_deleted BOOLEAN default false — never hard-delete records

SECTION 12 — CRITICAL TYPESCRIPT TYPES

```
<br/>// appointment.types.ts<br/><br/>export type AppointmentStatus =<br/> | 'DRAFT'  
| 'PENDING_APPROVAL' | 'APPROVED' | 'REJECTED'<br/> | 'SCHEDULED' | 'CONFIRMED' | '  
IN_PROGRESS'<br/> | 'ASSESSMENT_PENDING' | 'ASSESSMENT_COMPLETE'<br/> | 'REPORT_PEN  
DING' | 'REPORT_SUBMITTED' | 'REPORT_REVIEWED'<br/> | 'INVOICE_PENDING' | 'INVOICE_S  
ENT' | 'PAID'<br/> | 'OVERDUE' | 'ESCALATED' | 'ARCHIVED' | 'CANCELLED';<br/><br/>ex  
port type AppointmentType =<br/> | 'ORTHOPAEDIC' | 'PSYCHIATRIC' | 'NEUROLOGICAL' |  
'GENERAL_PRACTICE'<br/> | 'OCCUPATIONAL_THERAPY' | 'PHYSIOTHERAPY' | 'RADIOLOGICAL'  
| 'OTHER';<br/><br/>export interface AppointmentMasterRecord {<br/> id: string;<br/>  
reference_number: string; // KAM-YYYY-NNNN<br/> status: AppointmentStatus;<br/> a  
ppointment_type: AppointmentType;<br/> appointment_datetime: string;<br/> venue: st  
ring;<br/> duration_minutes: number;<br/> claimant: ClaimantSummary;<br/> attorney  
: AttorneySummary;<br/> expert: ExpertSummary | null;<br/> checklist: ChecklistItem  
[];<br/> assessment: Assessment | null;<br/> report: Report | null;<br/> finance:  
InvoiceRecord | null;<br/> communications: CommunicationEntry[];<br/> notes: Intern  
alNote[];<br/> timeline: TimelineEvent[];<br/> documents: Document[];<br/> created  
_at: string;<br/> updated_at: string;<br/>}<br/><br/>// user.types.ts<br/>export typ  
e UserRole = 'admin' | 'staff' | 'finance' | 'viewer'<br/> | 'attorney' | 'expert' |  
'claimant';<br/><br/>// finance.types.ts<br/>export type InvoiceStatus = 'DRAFT' | '  
SENT' | 'PAID'<br/> | 'OVERDUE' | 'ESCALATED' | 'VOIDED';<br/><br/>// All amounts in  
CENTS (integer). Display layer ÷100 → ZAR<br/>export interface Invoice {<br/> id:  
string;<br/> invoice_number: string;<br/> appointment_id: string;<br/> amount_cent  
s: number;<br/> vat_cents: number;<br/> total_cents: number;<br/> status: InvoiceS  
tatus;<br/> due_date: string;<br/> paid_at: string | null;<br/> payment_method: st  
ring | null;<br/>}<br/>
```

SECTION 13 — SECURITY & ROLE PERMISSIONS

13.1 Role Permission Matrix

Module / Action	admin	staff	finance	viewer
View appointments	✓	✓	✓	✓
Create appointment	✓	✓	✗	✗
Approve / Reject	✓	✓	✗	✗
Assign expert	✓	✓	✗	✗
View finance / invoices	✓	✗	✓	✗
Create / edit invoice	✓	✗	✓	✗
Record payment	✓	✗	✓	✗
Void invoice	✓	✗	✗	✗
View reports	✓	✓	✗	✓
Approve / review report	✓	✓	✗	✗
Manage staff	✓	✗	✗	✗
View audit logs	✓	✗	✗	✗
Edit system settings	✓	✗	✗	✗
View analytics	✓	✓	✓	✗
Manage CRM (attorneys/experts)	✓	✓	✗	✗
Export CSV data	✓	✓	✗	✗

13.2 Route Protection

- All /admin/* routes wrapped in component
- AdminRoute checks: (1) session exists, (2) role is admin or staff
- Finance-only routes additionally check role = 'admin' | 'finance'
- Unauthenticated → redirect to /login
- Wrong role → redirect to /unauthorised (custom 403 page)
- Session expired → redirect to /login with 'session-expired' query param

SECTION 14 — SHARED COMPONENTS SPECIFICATION

Every component listed below must be built before any page uses it:

StatCard

Props: title, value, subtitle?, trend? (up|down|neutral), onClick? | Renders a KPI card

DataTable

Props: columns, rows, loading, onClick, emptyStateText | Renders table with sort

SearchBar

Props: value, onChange, placeholder, onClear | Controlled input with clear button

FilterBar

Props: filters (array of FilterConfig), activeFilters, onChange | Renders filter pills

ModalWrapper

Props: isOpen, onClose, title, children, size (sm|md|lg|xl) | Accessible modal

ConfirmDialog

Props: isOpen, onConfirm, onCancel, message, danger? | Confirmation modal

SkeletonLoader

Props: rows?, type (table|card|list) | Animated loading placeholder

EmptyState

Props: title, message, actionLabel?, onAction? | Empty list/result state

ErrorCard

Props: message, onRetry? | Error display with optional retry

AppointmentStatusBadge

Props: status: AppointmentStatus | Colour-coded status pill

NotificationPanel

Slide-in panel, fetches user notifications, mark-read actions

SECTION 15 — COMMUNICATIONS CENTER

CommunicationsCenter.tsx gives admin/staff a unified view of all outbound communications sent by the platform. It is read-only (logs only). Sending new communications happens from appointment detail context.

- Filter by: type (email | SMS | OTP | system), entity, date range, status (sent | failed)
- Each entry shows: timestamp, type icon, recipient, subject/message, appointment reference, status
- Resend option for failed emails (admin only) — triggers re-send via Edge Function
- Email preview: clicking subject opens full email content in a modal

SECTION 16 — BUILD ORDER & FINAL CHECKLIST

Build in this exact order:

#	File / Module	Notes
1	Types	Define all TypeScript types in /types/ — no engine or component before types
2	auditEngine.ts	Build first — all other engines depend on it
3	notificationEngine.ts	Independent — build early
4	documentEngine.ts	Independent — build early
5	communicationEngine.ts	Depends on audit engine
6	financeEngine.ts	Depends on audit + communication engines
7	reportEngine.ts	Depends on audit + communication + document engines
8	workflowEngine.ts	Depends on all engines
9	appointmentEngine.ts	LAST engine — depends on ALL other engines
10	Service files	One service per engine — thin Supabase wrapper layer
11	Shared components	Build all shared/ui components before pages
12	Admin Layout	AdminLayout, AdminSidebar, AdminHeader
13	Dashboard Home	AdminDashboardHome.tsx
14	Appointment List	AppointmentManagement.tsx
15	Appointment Detail	AppointmentDetail.tsx — the master record view
16	CRM Modules	Attorneys, Experts, Claimants
17	Finance Dashboard	FinanceDashboard.tsx
18	Reports Dashboard	ReportsDashboard.tsx
19	Communications	CommunicationsCenter.tsx
20	Analytics	Analytics.tsx
21	Staff Management	StaffManagement.tsx
22	Audit Logs	AuditLogs.tsx
23	System Settings	SystemSettings.tsx

**END OF ADMIN DASHBOARD DEVELOPMENT PROMPT — KUTLWANO &
ASSOCIATES**

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